The Perils of Bilateral Sovereign Debt

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Official Sovereign Debt

- A large share of sovereign borrowing takes the form of official debt
  - … Multilaterals, development banks, other governments

- Emergence of new bilateral creditors **outside** the Paris Club IDS data
  - … with claims to seniority and sometimes **confidential** terms
  - … esp. when Central Banks are involved

Questions

- How does the presence of a large senior lender affect sovereign debt markets?
- What are its welfare implications for borrowing governments?

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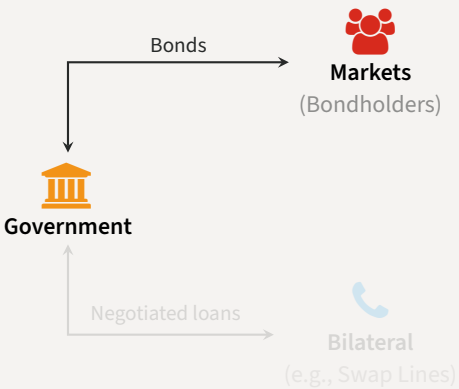
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Evaluating Senior Official Creditors

Quantitative sovereign debt model with

- Competitive creditors in private markets
- Large bilateral lender
  1. Superior enforcement [de-facto seniority]
  2. Bargained terms [price and quantity]
  3. Short-maturity loans
- Prime example: Central Bank swap lines (Horn et al., 2021)
- Focus on the interaction between both funding sources



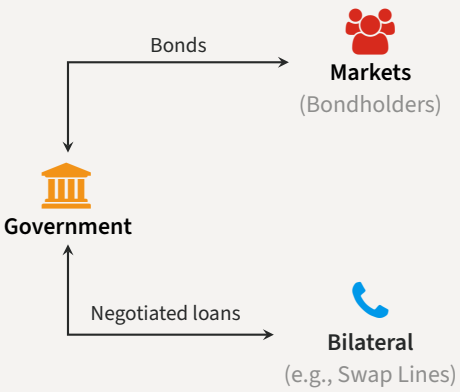
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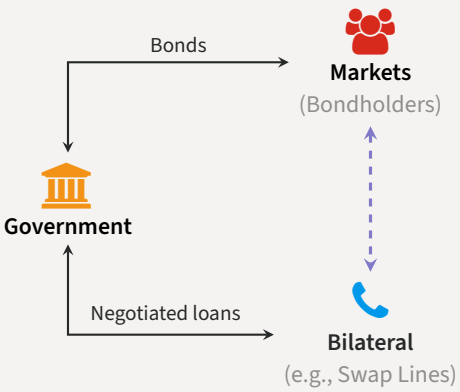


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Relational Overborrowing

Main findings

- Bilateral loans have significant effects on equilibrium outcomes
  - … provide funding when other sources dry up (e.g. because of default risk) ▲
  - … can also incentivize more **risk-taking** ▼

- If the rate on bilateral loans is decreasing in *market* debt [cross-elasticity]
  - … government issues market debt more quickly, delevers more slowly
  - … spends longer in the risky region
  - … defaults more frequently
  - … welfare losses for the government

- Cross-elasticity can emerge endogenously from bargaining
  - … market issuance raises cash-on-hand and strengthens the gov't negotiating stance
  - … **relational overborrowing**

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Literature

- Sovereign debt/default with interactions from ‘official’ debt
  - … senior debt (Hatchondo, Martinez & Önder 2017), senior debt with conditionality (Boz 2011, Fink & Scholl 2016), bailout agencies (Corsetti, Guimarães & Roubini 2006, Kirsch & Rühmkorf 2017, Roch & Uhlig 2018), official debt (Dellas & Niepelt 2016, Arellano & Barreto 2024, Liu, Liu & Yue 2025, Erce, Galli & Yurdagul 2026)

- Data on new official creditors
  - … Horn, Reinhart & Trebesch 2021a, 2021b, Gelpern et al. 2021, Horn, Parks, Reinhart & Trebesch 2023

- Central Bank swap lines
  - … among advanced economies (Bahaj & Reis 2021, Cesa-Bianchi, Eguren-Martin & Ferrero 2022), data for emerging-market borrowers (Perks, Rao, Shin & Tokuoka 2021)

Roadmap

- Theory
  - Bargaining and the Cross-Elasticity
- Quantitative Model
  - Endogenous Bargaining
  - Programming the Large Lender
- Concluding remarks

Theory

Theory

Bargaining and the Cross-Elasticity

Bargaining Induces a Cross-Elasticity

- Two periods $t \in \{0, 1\}$, endowments $y_0 < y_1$
- Government discount factor β, lender's discount factor $\beta_L > \beta$
- **Bargaining** transfer x at $t = 0$ for repayment $x(1 + r)$ at $t = 1$, weight θ for lender

$$\max_{x,r} \mathcal{L}(x, r)^{\theta} \times \mathcal{B}(x, r; y_0, y_1)^{1-\theta}$$

where

$$\mathcal{L}(x, r) = \underbrace{-x + \beta_L x(1 + r)}_{\text{agreement}}$$

$$\mathcal{B}(x, r; y_0, y_1) = \underbrace{u(y_0 + x) + \beta u(y_1 - x(1 + r))}_{\text{agreement}} - \underbrace{(u(y_0) + \beta u(y_1))}_{\text{threat point}}$$

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Bargaining Induces a Cross-Elasticity

- First-order conditions for bargaining problem

$$u'(y_0 + x) = \frac{\theta}{1-\theta} \frac{\mathcal{B}(x, r; y_0, y_1)}{\mathcal{L}(x, r)}$$

Surplus sharing

$$u'\underbrace{(y_0 + x)}_{c_0} = \frac{\beta}{\beta_L} u'\big(\underbrace{y_1 - x(1+r)}_{c_1}\big)$$

Euler

- Can prove that

$$\frac{\partial x}{\partial y_0} \leq 0 \qquad \text{and} \qquad \frac{\partial r}{\partial y_0} \leq 0$$

- **Later** y_0, y_1 are given by market debt dynamics, so
  - Recent issuances: increases y_0 and decreases y_1, makes you stronger
  - Legacy debt: lowers y_0 and y_1, typically makes you weaker

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$$\frac{\partial r}{\partial y_0} = -\frac{\theta\beta_L \frac{\partial \mathcal{B}}{\partial y_0}}{\theta\beta_L \frac{\partial \mathcal{B}}{\partial r} - (1-\theta)\beta \frac{\partial \mathcal{L}}{\partial r} u'(c_1) + (1-\theta)\beta\mathcal{L}u''(c_1)x} < 0$$

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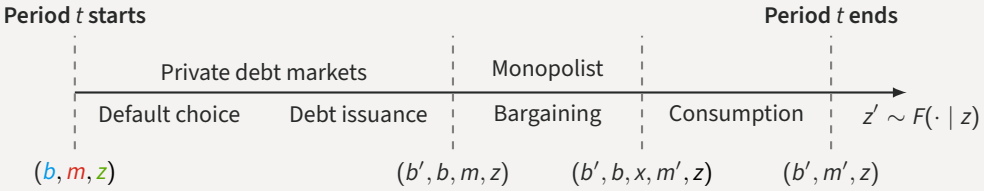
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Quantitative Model

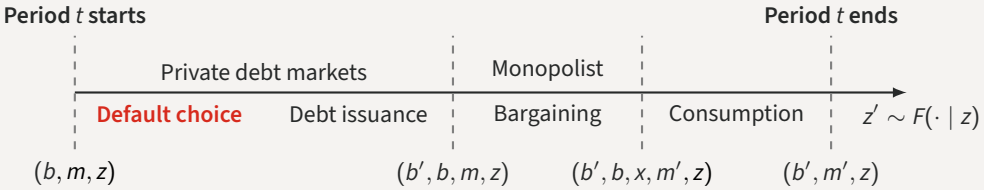
Timeline of Events

- Enter period t owing b to bondholders, m to monopolist, income $y(z)$



Timeline of Events

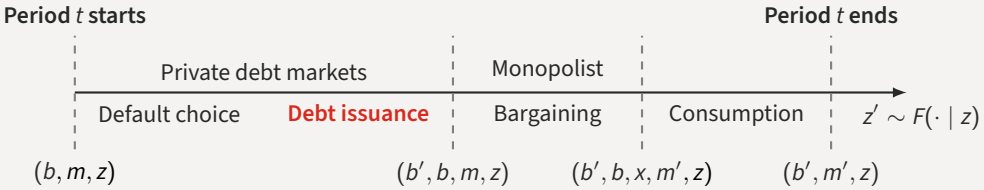
- Choose to repay or default the *market* debt subject to convex output costs



$$v(b, m, z) = \max \left\{ v_R(b, m, z) + \epsilon_R, v_D(m, z) + \epsilon_D \right\}$$

Timeline of Events

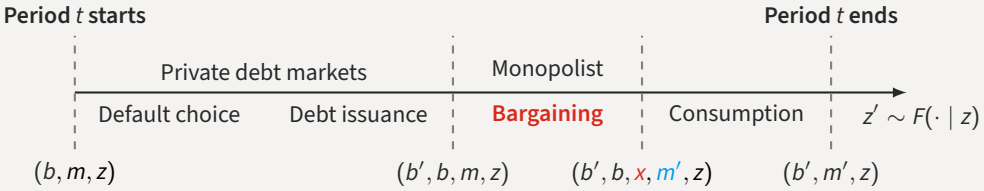
- If repaid, issue new (long-term) debt b' in markets at price q [coupon rate κ, maturity $1/\delta$]



$$q(b', b, m, z) = \beta_L \mathbb{E}\left[\left(1 - 1_{\mathcal{D}}(b', m', z')\right)\left(\kappa + (1-\delta) q(b'', b', m', z')\right) \mid z\right]$$

Timeline of Events

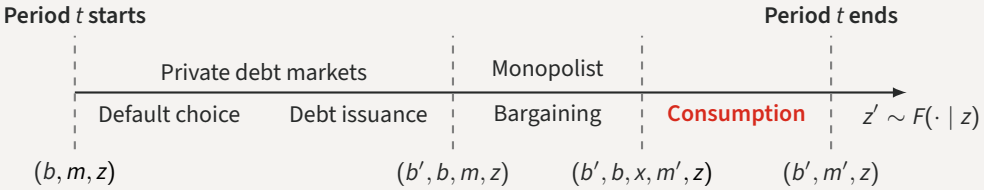
- Meet with senior lender, decide any transfers x and new/remaining balance m'



$$r(x, m', m) = \frac{m'}{x + m} - 1$$

Timeline of Events

- Consume output plus revenues from debt issuance plus transfers minus debt service



$$c_R = y(z) + q(b', b, m, z)(b' - (1 - \delta)b) + x_R(b', b, m, z) - \kappa b$$
$$c_D = y(z) - h(z) + x_D(m, z); \qquad x_D(m, z) \leq \Gamma(m)$$

Calibration

- Standard strategy in sovereign debt: match Argentina 1993-2001 to model w/o swaps

	Parameter	Value
Sov's discount factor	β	0.9607
Sov's risk aversion	γ	2
Pref. scale	χ	0.025
Lender's barg. power	θ	0.5
Risk-free rate	r	0.01
Duration of debt	ρ	0.05
Income autocorr.	ρ_z	0.9484
Std. dev. y_t	σ_z	0.02
Reentry prob.	ψ	0.0385
Default cost: linear	d_0	-0.2514
Default cost: quadratic	d_1	0.292

	Data	Only market
Avg spread (bps)	815	740
Std spread (bps)	443	485
Debt-to-GDP (%)	17.4	18.1

Note moments from pre-default samples

Quantitative Model

Endogenous Bargaining

Bargaining Stage with Senior Lender

- At state z, owing debt b bonds and m on the loan and having issued b'

$$\max_{x,m'} \mathcal{L}_R(b', x, m, m', z)^{\theta} \times \mathcal{B}_R(b', b, x, m, m', z)^{1-\theta}$$

Lender surplus

Government surplus

- Lender's surplus

$$\mathcal{L}_R(b', x, m, m', z) = \underbrace{\left(-x + \beta_L \mathbb{E}\left[h(b', m', z') \mid z\right]\right)}_{\text{agreement}} - \underbrace{\left(m + \beta_L \mathbb{E}\left[h(b', 0, z') \mid z\right]\right)}_{\text{threat point}}$$

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same sdf as markets

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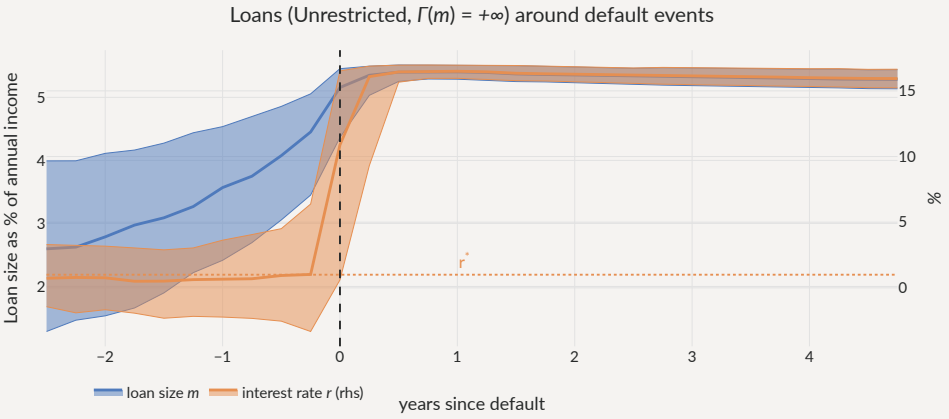
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Usage of Loans

- Sharp acceleration before defaults, maxed out *in* default if possible



Limiting Loans in Default

- **Limited**: cannot borrow in default $\Gamma(m) = m$ **Unavailable**: $\Gamma(m) = 0$

	Only market	**Unrestricted,** $\Gamma(m) = +\infty$	**Limited,** $\Gamma(m) = m$	**Unavailable,** $\Gamma(m) = 0$
Avg spread (bps)	630	1,398	1,209	650
Std spread (bps)	474	894	860	381
$\sigma(c)/\sigma(y)$ (%)	104	99	100	104
Debt to GDP (%)	16.4	17.6	17.6	18
Loan to GDP (%)	0	2.44	2.22	0.903
Loan spread (bps)	–	-298	-126	523
Corr. loan & spreads (%)	–	68.9	67.7	63.8
Default frequency (%)	5.03	9.55	9.04	5.28
Welfare gains (rep)	–	-0.095%	-0.038%	0.12%

Note Statistics from each ergodic distribution

Limiting Loans in Default

- Default rates: ↑ with loans (less with Limited/Unavailable)

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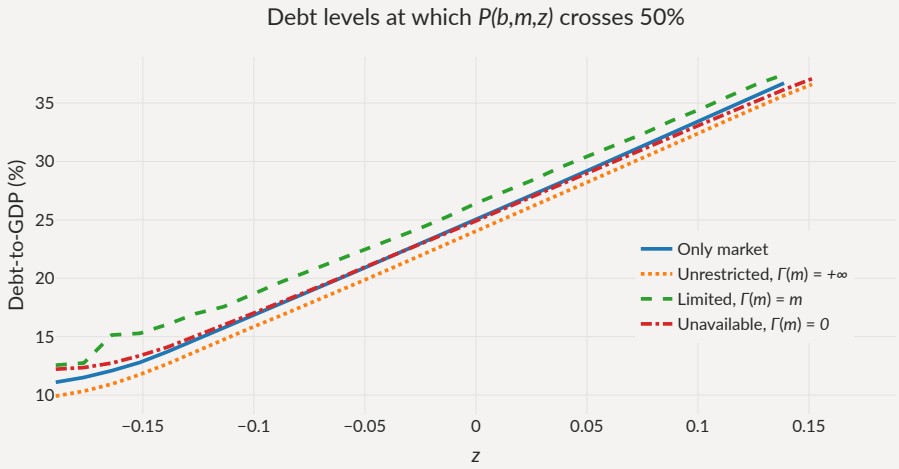
- Welfare: $\approx$ with loans (Unavailable $\succcurlyeq$ Limited $\succcurlyeq$ Unrestricted)

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Default Barriers with Loans

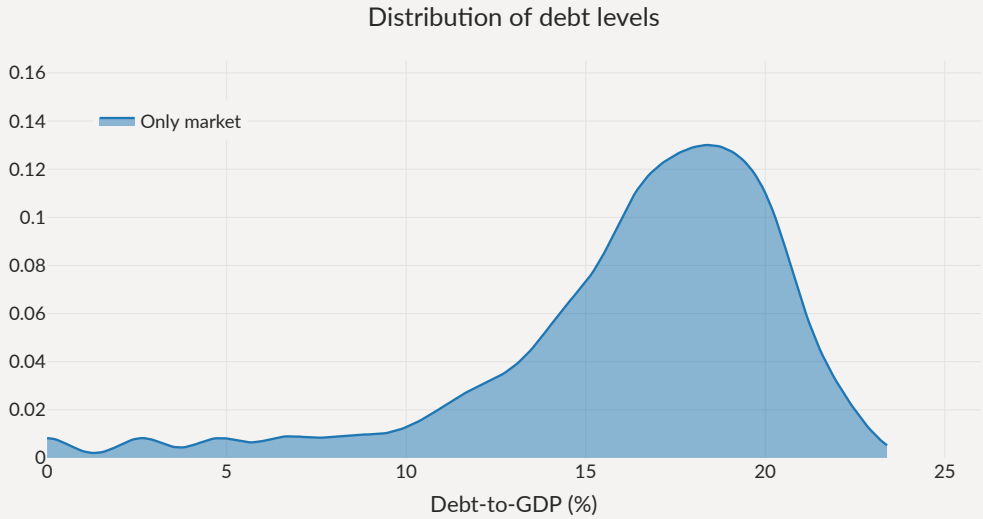
- Debt capacity changes marginally; does **not** explain the change in default frequency



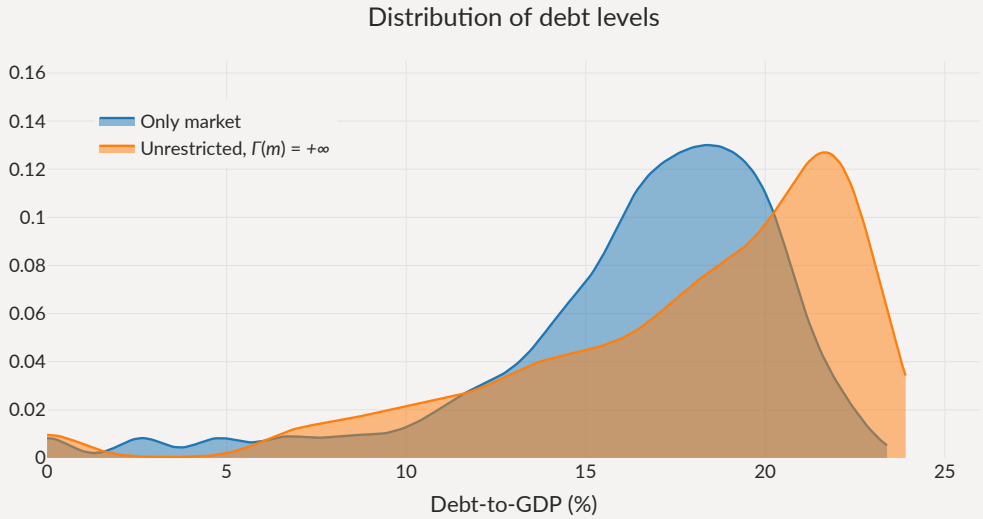
If Loans help repay the debt,

Why are there more defaults with loans?

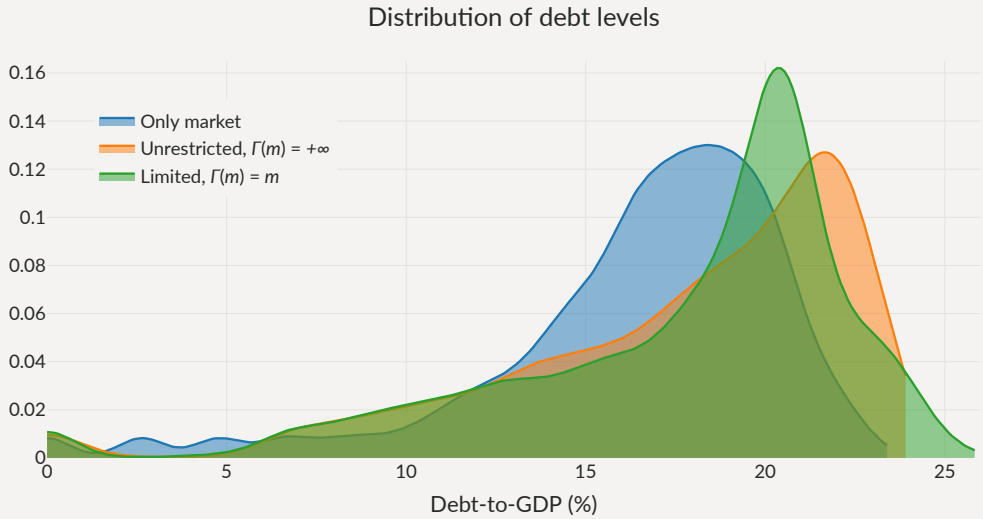
Debt Levels with Loans



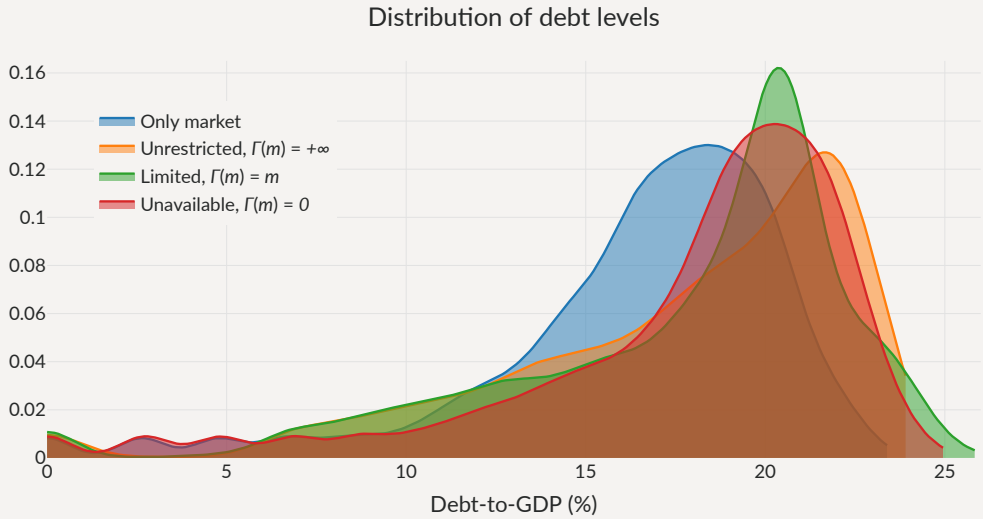
Debt Levels with Loans



Debt Levels with Loans



Debt Levels with Loans



Relational Overborrowing

Government's surplus

$$\mathcal{B}_R(b', b, x, m, m', z) = u\Big(\overbrace{y(z) + B(b', b, m, z) + x}^{=c_A}\Big) + \beta\mathbb{E}\left[v(b', m', z') \mid z\right] - \Big(u\Big(\underbrace{y(z) + B(b', b, m, z) - m}_{=c_T}\Big) + \beta\mathbb{E}\left[v(b', 0, z') \mid z\right]\Big)$$

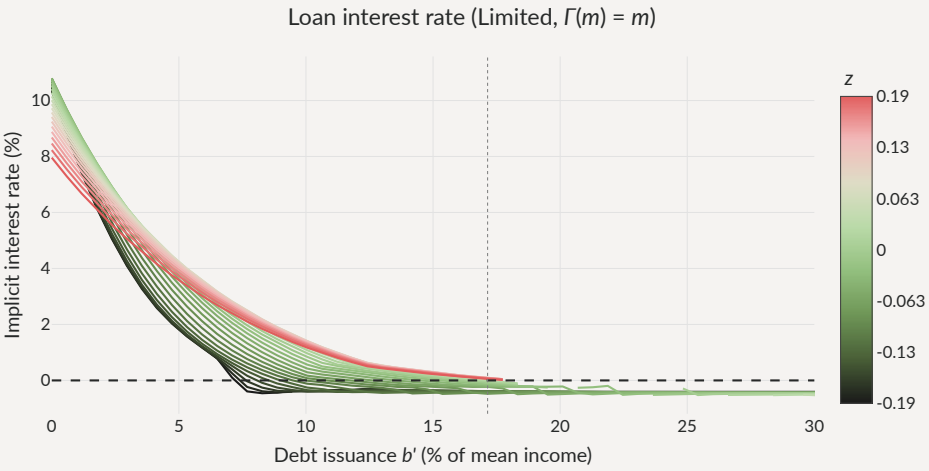
$$u\left(c_A\right) - u\left(c_T\right) \simeq u'(y(z) + B(b', b, m, z) - m) \cdot (x + m)$$

- Revenues from debt issuance $B(b', b, m, z)$ modulate the value of the threat point
  - After large revenues (high q, high b'), gov't flush with cash, **strong** in bargaining
  - After bad issuance (low q *or* low b'), gov't **weak** in bargaining

Risk-taking Incentives

Drawings

Threat point manipulation: increase market b' to reduce bilateral r **Cross-elasticity**



Government's Euler equation for bonds

$$u'(c)\left(q + \frac{\partial q}{\partial b'} i + \frac{1}{1+r}\frac{\partial m'}{\partial b'} + \frac{\partial \frac{1}{1+r_b}}{\partial b'} m'\right) = \beta \mathbb{E}\left[v_b(b', m', z') + v_m(b', m', z')\frac{\partial m'}{\partial b'} \mid z\right]$$

- An extra bond:
  - … yields marginal revenue $q + \frac{\partial q}{\partial b'} i$ [standard debt-Laffer curve effect]
  - … makes you **stronger** in bargaining now
    - … affects the amount m' borrowed from senior lender
    - … affects terms of borrowing with senior lender
  - … makes you **weaker** in bargaining later

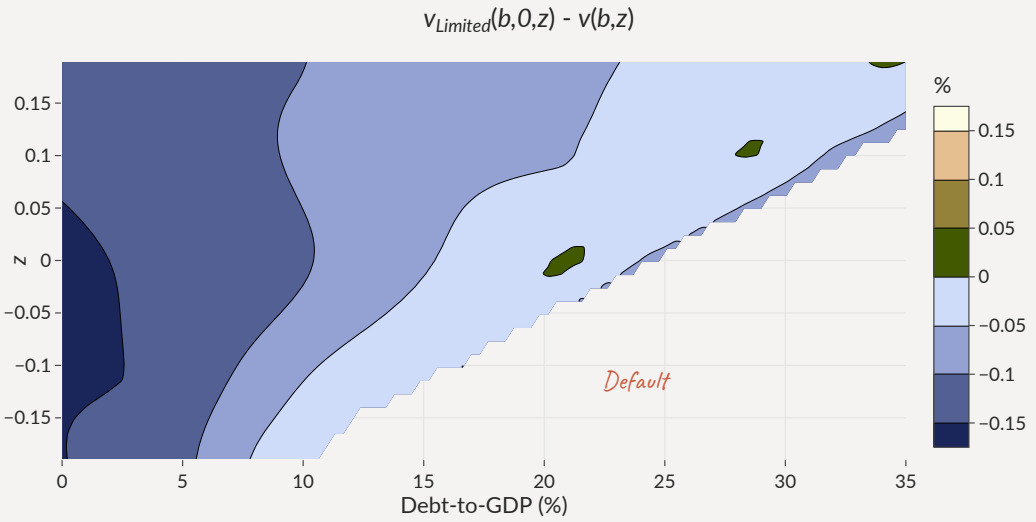
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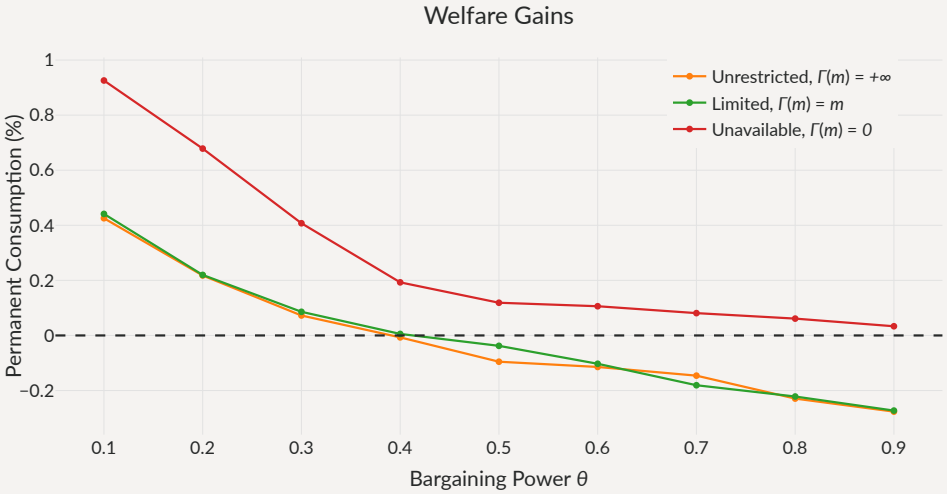
Welfare Effects of Bilateral Loans

Welfare gains overall near barrier: postponing default dominates



Bargaining Power and Welfare

Welfare gains dissipate quickly as the senior lender's bargaining weight rises



Quantitative Model

Programming the Large Lender

Possible Rules

- Explore interest rate rules of the form

$$r(b', m') = \max\{r^\star, \alpha_0 + \alpha_b b' + \alpha_m m'\}$$

- Three versions

Size-dependent
$\alpha_0 > 0, \alpha_b = 0, \alpha_m > 0$
(no cross-elasticity)

Risk-inducing
$\alpha_0 > 0, \alpha_b < 0, \alpha_m \geq 0$
(mimics eq'm w. bargaining)

Optimal
(for borrower welfare)
$\alpha_0 > 0, \alpha_b > 0, \alpha_m > 0$

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Equilibrium with Exogenous Rules

- 'Only market' standard calibration to Argentina 1993-2001

	Only market	Size dependent r	Risk inducing r	Optimal r
Avg spread (bps)	630	413	831	227
Std spread (bps)	474	240	447	130
$\sigma(c)/\sigma(y)$ (%)	104	105	104	106
Debt to GDP (%)	16.4	17.3	17.1	17.7
Loan to GDP (%)	0	0.727	0.486	2.56
Loan spread (bps)	–	662	296	516
Corr. loan & spreads (%)	–	49.4	-29.2	38.9
Default frequency (%)	5.03	3.2	6.19	1.92
Welfare gains (rep)	–	0.15%	-0.097%	0.69%

Note Statistics from each ergodic distribution

Equilibrium with Exogenous Rules

- Default rates: ↓ with size dependent ↑ with risk-inducing

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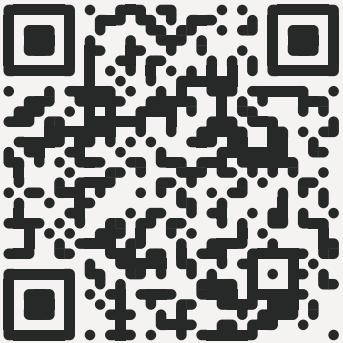
Concluding remarks

The Perils of Bilateral Sovereign Debt

- Simple model of borrowing from markets and a senior bilateral lender
  - … strong interaction between two markets for sovereign debt
  - … even if bilateral loans are **not** used intensely on the equilibrium path

- **Dangerous** when bilateral interest rate responds negatively to *market* debt
  - … the cross-elasticity erodes the disciplining role of spreads
  - … sequential bargaining is a plausible source of the cross-elasticity

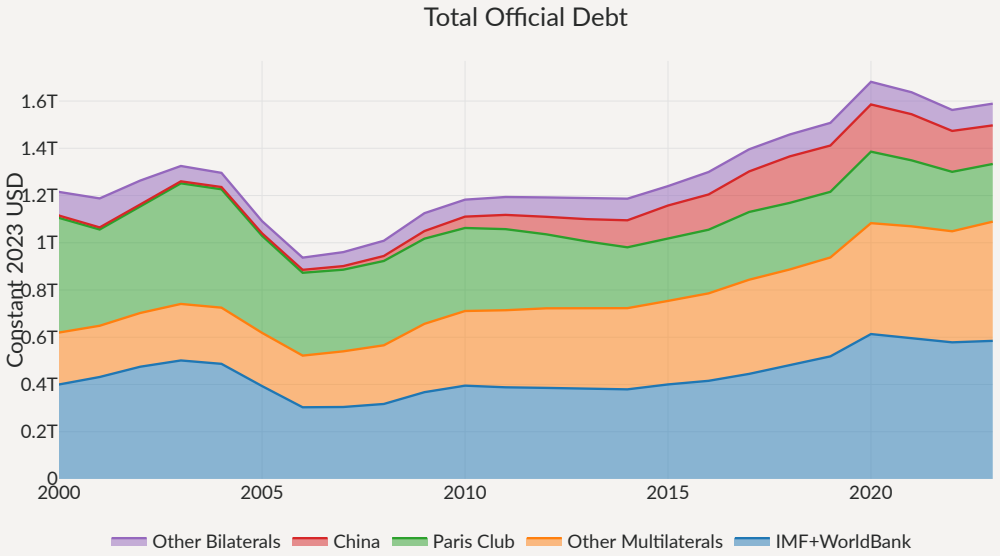
- Transparent, pre-specified **rules** can reverse the force
  - … rules with opposite cross-elasticity improve welfare without losses for lender
  - … **Practical diagnostic:** ask how bilateral rates respond to market issuance or spreads



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A New Landscape for Official Sovereign Debt

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Risk-taking Incentives

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